

1. Introduction

The rapid advancement of information and communication technologies has transformed how higher educational institutions create, store, and manage information. Academic departments generate large volumes of records such as lecture materials, examination papers, score sheets, research documents, and administrative files, all of which are essential for teaching, learning, and institutional accountability. As these records continue to increase, the need for efficient, secure, and scalable file storage systems has become a major concern for universities and technical institutions (Armbrust et al., 2010).

Traditionally, many academic departments rely on physical storage methods and localized digital storage solutions such as filing cabinets, personal computers, external drives, and on-premise servers. Physical examination papers, in particular, occupy significant office space and are prone to deterioration, loss, and misplacement. Retrieving old examination papers is often time-consuming and inefficient, while localized digital storage systems suffer from limited accessibility, poor scalability, and high risk of data loss due to hardware failure (Mell & Grance, 2011).

Cloud computing has emerged as a viable solution to these challenges by providing on-demand access to shared computing resources over the internet. Cloud-based file storage systems enable centralized storage, scalability, automated backup, and secure access to files from different locations and devices. These features make cloud storage particularly suitable for managing academic records and examination documents within educational institutions (Zhang et al., 2010).

At Koforidua Technical University, particularly within the Computing Sciences Department, the management of large volumes of physical examination papers has become increasingly challenging. These documents consume valuable physical space and are difficult to retrieve when needed. In addition, there is no automated mechanism to remind lecturers to upload examination scores or to control access to sensitive academic records. This situation highlights the need for a cloud-based file storage system that supports digital archiving, file size optimization, automated alerts, and controlled access.

This study therefore focuses on the design and development of a cloud-based file storage system for the Computing Sciences Department of Koforidua Technical University. The proposed system aims to digitize and securely store examination papers, reduce the size of scanned PDF files, notify lecturers on timely upload of examination scores, and provide controlled, time-based access to authorized users, thereby improving efficiency, security, and sustainability in academic record management.

2. Problem Statement

The Computing Sciences Department of Koforidua Technical University faces significant challenges in managing a growing volume of physical examination papers and academic records. These documents occupy substantial physical space and are stored using manual filing systems that make organization and long-term preservation difficult. Retrieving specific examination papers, especially older records, is time-consuming and inefficient, often delaying academic and administrative processes. Additionally, the absence of a centralized digital system limits secure access, increases the risk of document loss or damage,

and provides no automated mechanism to remind lecturers to upload examination scores within required timelines. These challenges highlight the need for a cloud-based file storage system that supports digitization, efficient storage, controlled access, and timely academic record submission.

3. Objectives

General Objective

To design and develop a secure, scalable, and efficient cloud-based file storage system that supports digital archiving, controlled access, and effective management of examination records within the Computing Sciences Department of Koforidua Technical University.

Specific Objectives

1. To design and implement a file compression model that reduces the size of scanned PDF examination documents before uploading to the cloud.
2. To develop a cloud-based storage system that enables secure scanning, uploading, retrieval, and long-term preservation of examination papers.
3. To design and implement an automated alert model that notifies lecturers to upload examination scores after assessment recording.
4. To develop an access control model that allows lecturers and administrators to grant access to specific users within defined time periods.

4. Relevance / Significance of the Study

The proposed system will significantly improve examination record management within the Computing Sciences Department by reducing physical storage requirements and enhancing retrieval efficiency. The integration of PDF compression will optimize cloud storage usage, while automated alerts will promote timely submission of examination scores. Controlled access mechanisms will improve data security and accountability. The study will also contribute practical knowledge to cloud application development and digital archiving in educational institutions.

5. Scope of the Study

This study is limited to the design and development of a cloud-based file storage system for the Computing Sciences Department of Koforidua Technical University. The system will focus on scanning and storing examination papers, PDF size reduction, automated lecturer notifications, user authentication, and time-based access control. The study will not cover university-wide deployment or advanced analytics features.

6. Limitations of the Study

The study may be constrained by internet connectivity challenges, limited time for development and testing, and dependency on selected cloud service providers. The effectiveness of the system will also depend on user compliance and availability of scanning resources.

7. Methodology

The study will adopt a software development life cycle approach comprising system analysis, design, implementation, testing, and deployment. Data will be collected through observation and interviews with departmental staff to gather system requirements. The system will be developed using modern web technologies and deployed on a cloud platform. Functional and usability testing will be conducted to validate system performance.

8. Literature Review

The literature review will examine existing studies on cloud computing, digital document archiving, cloud-based file storage systems, PDF compression techniques, notification systems, and access control mechanisms in cloud environments. Emphasis will be placed on prior implementations in educational institutions and best practices for secure academic record management.

9. Conclusion

This proposal presents a comprehensive plan to design and develop a cloud-based file storage system for managing examination papers and academic records in the Computing Sciences Department of Koforidua Technical University. By incorporating digital archiving, file compression, automated alerts, and controlled access models, the proposed system aims to enhance efficiency, security, and sustainability in academic record management.